

PRD: AI Competitor Intelligence Pipeline

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1. Project Overview

The **AI Competitor Intelligence Pipeline** is a Product Management automation workflow that helps track competitor pricing and feature information from live web pages. Instead of manually checking competitor pricing pages, the workflow scrapes a competitor URL, extracts pricing and feature details using AI, formats the data into a structured API payload, and stores the results in Airtable.

The MVP was tested using the Notion pricing page and successfully created structured Airtable records for multiple pricing plans.

2. Problem Statement

Product Managers need to regularly monitor competitor pricing, packaging, and feature positioning. This process is often manual, repetitive, and inconsistent.

- Pricing or feature changes can be missed.
- Data is copied inconsistently into spreadsheets.
- Research takes time away from higher-value PM work.
- Competitor insights are not always structured for comparison.
- Teams may not know when competitor data was last checked.

3. Target Users

User Type	Description
Primary user	Product Manager responsible for competitor research, pricing analysis, and roadmap planning.
Secondary users	Product Marketing Manager, Sales Enablement team, Strategy or Growth team, and founders or early-stage product teams.

4. Goals

The goal is to automate the first layer of competitor intelligence collection. The workflow should:

- Accept a competitor pricing page URL.
- Scrape live website content.
- Extract pricing plans, features, target users, and PM insights.
- Format the data into an Airtable-compatible API payload.
- Store each pricing plan as a row in Airtable.
- Reduce manual effort in competitor research.

5. Non-Goals

This MVP does not include:

- Automatic weekly scheduling.

- Multi-competitor batch processing.
- Historical change detection.
- Automated Slack or email alerts.
- Full dashboard reporting.
- Guaranteed extraction accuracy for every website layout.
- Scraping pages behind login or paywalls.

6. MVP Scope

Area	Included in MVP	Not included in MVP
Input	Single competitor URL input.	Bulk competitor URL upload.
Extraction	Website scraping and AI extraction of pricing fields.	Guaranteed extraction for all dynamic websites.
Storage	Airtable records, one row per pricing plan.	Advanced dashboards and reporting.
API testing	Webhook.site used to validate POST payloads.	Permanent Webhook.site dependency.
Automation	Manual run from Gumloop.	Scheduled recurring runs.

7. User Stories

- 1 As a Product Manager, I want to enter a competitor pricing URL so that I can collect pricing data without manually copying it.
- 2 As a Product Manager, I want the workflow to extract pricing plans and key features so that I can compare competitor offerings faster.
- 3 As a Product Manager, I want the extracted data stored in Airtable so that I can review and organize competitor intelligence in a structured table.
- 4 As a Product Manager, I want each record to include a PM insight so that raw pricing data is connected to product strategy.

8. Functional Requirements

ID	Requirement	Acceptance Criteria
FR1	Competitor URL Input	Workflow accepts a pricing page URL such as https://www.notion.com/pricing .
FR2	Website Scraping	Scraper returns readable pricing page content including plan and feature details.
FR3	AI Data Extraction	Workflow extracts Competitor Name, Plan Name, Price, Billing Period, Target User, Key Features, and PM Insight.
FR4	JSON Formatting	Ask AI returns a valid Airtable JSON body with a fields object and exact Airtable field names.
FR5	Airtable API Integration	Call API creates one Airtable row per pricing plan and returns a record ID beginning with rec.
FR6	API Validation	Webhook.site is used to validate POST payloads before the Airtable connection is finalized.

9. Airtable Data Model

The Airtable table is named **Pricing Records**. Each row represents one competitor pricing plan.

Field Name	Description
Competitor Name	Name of the company or product.
Plan Name	Name of the pricing plan.
Price	Listed price or custom pricing.
Billing Period	Monthly, yearly, per user, per seat, or similar billing detail.
Target User	Intended customer segment.
Key Features	Main features included in the plan.
PM Insight	Product Management interpretation of the plan.
Source URL	Pricing page used as the source.
Date Checked	Date the workflow was run.

10. Success Metrics

The MVP is successful if:

- The workflow accepts a competitor pricing URL.
- The scraper returns usable website content.
- AI extraction identifies multiple pricing plans.
- The JSON payload matches Airtable field names correctly.
- Airtable creates one row per pricing plan.
- The final table contains usable competitor pricing records.

Example success result:

```
Notion | Free      | $0
Notion | Plus       | $10
Notion | Business    | $20
Notion | Enterprise  | Custom pricing
```

11. Risks and Limitations

Risk	Impact	Mitigation
Pricing pages have different layouts	AI may miss or misread details.	Test with multiple competitor pages.
Website content is dynamic	Scraper may not capture all details.	Use advanced scraping if needed.
AI extraction may be imperfect	Incorrect records may enter Airtable.	Add review step or confidence score later.
Airtable field names must match exactly	API request may fail.	Keep schema documented.
API tokens may be exposed in screenshots	Security risk.	Blur all tokens and private URLs.

12. Future Enhancements

- Add multiple competitor URLs.
- Schedule weekly runs.

- Add change detection against previous records.
- Flag pricing changes automatically.
- Send Slack or email alerts.
- Add confidence scores for extracted fields.
- Build an Airtable dashboard.
- Add raw scraped text for auditability.
- Add competitor category or product segment fields.

13. Final Outcome

The MVP successfully demonstrates a working Product Management automation pipeline that turns competitor pricing pages into structured product intelligence. It shows practical PM skills in competitor research, pricing analysis, workflow automation, AI-assisted data extraction, API integration, and structured product documentation.